

PriceOS Pricing Engine - Completion Checklist

Summary: 28 complete, 27 pending (see legend below).

Legend: [x] = complete [] = pending

Implementation rows reflect a full-repo audit of Original_priceos/priceos. Re-run this script after shipping features; edit CHECKLIST tuples to adjust.

Strategy & planning (documentation)

- [x] Competitive analysis: PriceLabs pricing stack documented (base -> seasonal -> DOW -> lead time -> demand -> occupancy -> gap -> adjacent -> LOS -> guardrails)
- [x] Customization map: all PriceLabs-style levers categorized into Tier 1 / 2 / 3 with revenue-impact framing
- [x] Prioritization: Dubai STR-focused scoring (80/20: top 6 Tier 1 items ~80% impact)
- [x] Roadmap: Phases 1-4 with sprint mapping (Foundation -> Intelligence -> Advanced -> V2)
- [x] Differentiation narrative captured (agents, simulation-first, Dubai context, NL control, auto ADR)
- [x] Engineering backlog: sprint-board tickets exported for pricing engine workstreams (see Sprint Board folder)
- [x] Source document: PriceOS Pricing Engine Customizations (strategy PDF, Apr 2026)
- [x] Code audit: Original_priceos pricing engine (waterfall.ts, pipeline.ts, models, UI) - Apr 2026

Tier 1 - Must build (product/engine)

- [x] T1.1 Base price + min/max guardrails (engine: listing base + floor/ceiling clamp in waterfall)
- [] T1.1b Auto base from comp ADR into engine default (benchmark exists; not wired into computeDay base)
- [x] T1.2 Seasonal profiles (SEASON rules: date range, DOW mask, price/% adj, min/max overrides)
- [x] T1.3a Last-minute discount (flat % within days-out window; floor enforced via final clamp)
- [] T1.3b Last-minute gradual ramp (min/max discount over ramp days; Listing fields not used in waterfall)
- [] T1.4 Occupancy-based adjustments in rules engine (schema + Rules Studio UI; not applied in computeDay)
- [x] T1.5 Orphan / gap pricing (gap prevention + gap-fill discount by length band; not weekday split rules)
- [x] T1.6 Day-of-week pricing (configurable dowDays e.g. Thu/Fri; no auto-pattern from booking history)

Tier 2 - Should build

- [x] T2.7 Far-out premium (markup when lead time >= farOutDaysOut)
- [] T2.8 Minimum weekend pricing (weekendMinPrice on model/UI; not enforced in waterfall clamp)
- [x] T2.9 Date-specific overrides (listing-level rules: SEASON/EVENT/ADMIN_BLOCK, min stay, block, CTA/CTD)
- [] T2.9b Overrides at group / account scope (listing-level only today)
- [] T2.10 Adjacent booking factor (turnaround premium/discount)
- [] T2.11 Length-of-stay discounts applied to nightly rate (LOS_DISCOUNT rules only noted in engine note)

Tier 3 - Build later

- [] T3.12 Demand factor sensitivity / hotel weight
- [] T3.13 Customization hierarchy (account > group > sub-group > listing)
- [x] T3.14 Check-in / check-out day restrictions (allowedCheckinDays / allowedCheckoutDays in waterfall)
- [] T3.15 Channel-specific pricing offsets
- [] T3.16 Intra-day portfolio occupancy adjustments

Phase 1 - Foundation (Sprint 1-2)

- [] P1.1 Base + guardrails + auto-ADR as engine default (ADR via market flow; not pipeline base)
- [x] P1.2 Seasonal profiles engine (PricingRule SEASON + runPipeline)
- [x] P1.3 Day-of-week adjustments (UAE-style weekend days configurable)

- [x] P1.4 Date-specific override system + UI (rules API + Pricing Rules Studio)
- [] P1.5 Bar-level rate grid (not found as product surface)

Phase 2 - Intelligence layer (Sprint 3-4)

- [] P2.1 Last-minute curve + velocity awareness (flat discount only; no velocity in waterfall)
- [] P2.2 Occupancy in waterfall + portfolio-level pricing rules
- [x] P2.3 Orphan/gap pricing (implemented; not full multi-rule matrix / weekday split)
- [x] P2.4 Far-out minimum protection (implemented as far-out markup pass)
- [] P2.5 Minimum weekend pricing (field exists; not in computeDay)

Phase 3 - Advanced (Sprint 5-6)

- [] P3.1 Adjacent booking factor
- [] P3.2 Tiered LOS discounts in nightly calculation
- [x] P3.3 Competitor rate comparison UI (benchmark panel; no auto-positioning in engine)
- [] P3.3b Auto-positioning from comps in engine
- [] P3.4 Portfolio occupancy + booking-window segmentation in engine
- [] P3.5 Customization hierarchy

Phase 4 - V2 / post-launch

- [] P4.1 Intra-day pricing adjustments
- [] P4.2 Channel-specific offsets
- [] P4.3 Hotel weighting / demand sensitivity
- [] P4.4 CI/CO restrictions (already in V1 engine; phased doc item)
- [] P4.5 Automatic monthly base recalibration + nudges

Supporting UX / platform (sprint backlog)

- [x] Pricing calculation pipeline with layer stacking (4-pass waterfall in waterfall.ts)
- [] 365-day calendar grid with prices per cell (engine runs 365d; UI uses proposal table + 30d heatmap)
- [x] User overrides on automated prices (approve/reject; bulk-modify / bulk-save proposals APIs)
- [x] Competitor rate comparison panel (benchmark-panel + benchmark data)
- [x] Portfolio-level views (dashboard occupancy, market page; chat portfolio summary)
- [x] Market / comp workflow (market-setup + benchmark agents; recommended rates in DB/chat context)
- [x] Occupancy-based sorting (e.g. overview-client property list by occupancy)

Ticks from Original_priceos codebase audit + strategy PDF. Script: generate_pricing_engine_checklist_pdf.py